FOOD SCIENCE AND HUMAN NUTRITION

The Department of Food Science and Human Nutrition is jointly administered by the College of Agriculture and Life Sciences and the College of Health and Human Sciences. All curricula offered by the department are available to students in either college. These majors include:

- Culinary food science
- Dietetics
- Diet and exercise
- Food science
- Nutritional science
- Nursing

Visit the department web site at: www.fshn.hs.iastate.edu (http://www.fshn.hs.iastate.edu).

Undergraduate Study

Culinary Food Science

Culinary Food Science is an interdisciplinary degree that combines food science with culinary skills development, preparing students to become leaders in food product development. Through food science coursework, hands-on culinary training, and food industry internships, graduates are equipped to develop new innovative food products and optimize existing ones for improved taste, nutrition, and consumer appeal. In the food industry, culinary food scientists work in research and development of food products, culinary innovation, test kitchens, recipe development, marketing and sales, food regulation, and quality assurance.

The program is an approved Culinology® curriculum by the Research Chefs Association. Culinary food science graduates are prepared for careers as food product developers, food innovation managers, culinary application scientists, test kitchen managers, recipe developers, and food marketing and sales managers.

Dietetics

The Didactic Program in Dietetics (DPD) is accredited by the Accreditation Council for Education in Nutrition and Dietetics, the accrediting agency of the Academy of Nutrition and Dietetics. The dietetics undergraduate curriculum meets the academic requirements as the DPD. Additionally, the curriculum for concurrent bachelor's and master's degrees in diet and exercise meets the academic requirements of the DPD. Graduates of the program are eligible to apply for admission to accredited dietetics internships/supervised practice programs. Upon successful completion of the experience program and a master's degree, graduates are eligible to take the national examination administered by the Commission on Dietetic Registration to become a Registered Dietitian Nutritionist (RDN) and to practice in the field of dietetics. There is a $30 fee for a statement of verification of completion of the DPD.

The dietetics program includes study in basic sciences, nutrition, and food science with applications to medical dietetics, nutrition counseling and education, and community nutrition. Foodservice management is also an important aspect of the program. Graduates work in clinical settings, consulting, food companies, food services, sports or athletic programs, corporate wellness programs, care facilities for patients from neonatal to geriatric, and community or school health programs.

Diet and Exercise

A program for concurrent Bachelor of Science and Master of Science (B.S./M.S.) degrees in diet and exercise (https://fshn.hs.iastate.edu/find-your-major/diet-and-exercise/) is available. The program is jointly administered by the Department of Food Science and Human Nutrition (FSHN), within the College of Agriculture and Life Sciences and College of Health and Human Sciences, and the Department of Kinesiology within the College of Health and Human Sciences. Students interested in this program enroll as pre-diet and exercise students. In the third year, students apply for admission to the BS/MS program. Students not accepted into the program can continue toward completion of the BS degree in dietetics or kinesiology and health. Coursework has been designed to facilitate a 4-year graduation date for those students not accepted into the program and electing to complete a single undergraduate degree. Students accepted into the program will progress toward completion of B.S./M.S. degrees in diet and exercise.

Food Science

Food science is a discipline in which the principles of biological and physical sciences are used to study the nature of foods, the causes of their deterioration, and the principles underlying the processing, preparation, and safe handling of food. Food Science involves the application of science and technology to help ensure a safe, wholesome, and nutritious food supply. Biotechnology and toxicology interrelate with food science in the area of food safety. In the food industry, food scientists work in research and development of products or processes, production supervision, quality control, marketing and sales, test kitchens and recipe development, product promotion and communication. Food scientists also work in government regulatory agencies and academic institutions.

The food science major is approved by the Institute of Food Technologists, the national professional organization of food science. Career options include quality control/assurance; production supervision; management and sales; research careers in the food industry, government, or academia; business; journalism; food product formulation and recipe development; food promotion and communication; and consumer services in government and industry.

Students in food science have the opportunity to pursue a Master of Business Administration (http://www.fshn.hs.iastate.edu/undergraduate-programs/food-science/) (MBA) concurrently with the Bachelor of Science (B.S.) degree in food science. The program is designed so students can earn both the B.S. in food science and MBA in five years, to meet the needs of students who are interested in management careers in the food industry. Students apply for admission to the MBA program in the spring of the third year. The program for concurrent B.S. in food science/MBA degrees is a rigorous 5-year program, and admission is very selective.

Nutritional Science

Nutritional science looks at the connection between diet and health. Students learn how diet can play a crucial role in the cause, treatment, and prevention of many diseases. The pre-health and research focus of the coursework prepares students for work in research laboratories, graduate study in nutrition or biological sciences, or entrance into health professional programs, such as medical, dental, physician assistant, and pharmacy schools. Students gain a strong science education along with human nutrition expertise.

Nursing

The Bachelor of Science in Nursing (BSN) program at Iowa State University is a RN-to-BSN program, designed for those who have obtained an associate degree in nursing or are eligible for NCLEX, and desire to further their nursing career and education to the next level. Iowa State's RN-to-BSN program provides interactive learning opportunities where students can apply their real-world experiences and education to inspire innovation in their places of care. RN-to-BSN students will acquire knowledge, skills, and experience in five essential areas: research, leadership, population health, self-care, and health promotion. The program emphasizes providing professional, cultural, and ethically congruent care the respects the dignity and uniqueness of individuals and groups in diverse populations and locations.

The baccalaureate nursing program at Iowa State University of Science and Technology located in Ames, Iowa is approved by the:

Iowa Board of Nursing (IBON)
6200 Park Ave #100
Des Moines, IA 50321

The baccalaureate nursing program at Iowa State University of Science and Technology located in Ames, Iowa is accredited by the:

Accreditation Commission for Education in Nursing (ACEN)
3390 Peachtree Road NE, Suite 1400
Atlanta, GA 30326
(404) 975-5000

The most recent accreditation decision made by the ACEN Board of Commissioners for the baccalaureate nursing program is initial accreditation.

View the public information disclosed by the ACEN regarding this program at https://www.acenursing.org/acen-programs-05202024/iowa-state-university-of-science-and-technology.

For more information and RN-to-BSN student learning outcomes: https://nursing.iastate.edu/rn-bsn-program

FSHN Departmental Learning Outcomes

Upon graduation, students should be able to:

- Communicate effectively in their field of study using written, oral, visual and/or electronic forms.
- Demonstrate proficiency in ethical data collection and interpretation, literature review and citation, critical thinking and problem solving.
- Participate effectively in a group or team.
- Integrate creativity, innovation, or entrepreneurship in ways that produce value.
- Explain how human activities impact the natural environment and how societies are affected.
- Meet program specific learning outcomes.

For more information: https://fshn.hs.iastate.edu/staff-and-faculty/resources/outcomes-assessment/learning-outcomes/.

Communication Proficiency is certified by a grade of C or better in 6 credits of coursework in composition (ENGL 1500 Critical Thinking and Communication and ENGL 2500 Written, Oral, Visual, and Electronic Composition or other communication-intensive courses) and a grade of C or better in 3 credits of coursework in oral communication.

Minors - Undergraduate

The department offers minors in:

- culinary food science
- food and society
- food safety (interdepartmental minor)
- food science
- nutrition

All minors require at least 15 credits, including at least 9 credits in courses numbered 2000 or above, of which at least 6 credits in courses numbered 3000 or above, with at least 3 credits taken at Iowa State University. The minor must include at least 3 credits that are not used to meet any other department, college, or university requirement.

Prerequisites: Students must complete prerequisite requirements for courses included in the minor.

Minor in Culinary Food Science

FSHN 1040	Introduction to Culinary Skills	2
FSHN 1150	Food Preparation Laboratory	1-2
or FSHN 2150	Advanced Food Preparation Laboratory	
FSHN 2140	Scientific Study of Food	3
or FSHN 2070	Processing of Foods: Basic Principles and Applications	
FSHN 4930	Food Preparation Workshop	1
Select additional credits from the following list for a minimum of 15 credits for the minor:		
FSHN 2200	American Food and Culture	3
FSHN 3800	Food Production Management	3
FSHN 3800L	Food Production Management Experience	3
FSHN 4110	Food Ingredient Interactions and Formulations	2
FSHN 4910D	Supervised Work Experience: Culinary Science	1-4
FSHN 4930	Food Preparation Workshop (One Food Preparation Workshop is required for minor. Option to select one additional workshop.)	1
HSPM 1330	Food Safety Certification	1
FSHN 4820X	Global Cuisines	3

Minor in Food and Society

FSHN 1670	Introductory Human Nutrition and Health	3
FSHN 2420	The US Food System	3
FSHN 4700	Sustainable and Healthy Eating Patterns	3
Select 6 additional credits from:		6
FSHN 2200	American Food and Culture	
FSHN 3420	World Food Issues: Past and Present	
FSHN 3650	Obesity and Health	
FSHN 4420	Food, Culture, and Sustainability	
FSHN 4600	Global Nutrition, Health and Sustainability	
FSHN 4630	Community Nutrition and Health	
SOC 2200	Globalization and Sustainability	
SOC 4440	Sociology of Food and Agricultural Systems	

Interdepartmental Minor in Food Safety (Students majoring in Food Science or Culinary Food Science cannot declare a minor in Food Safety.)

FSHN 1010	Food and the Consumer	3
HSPM 1330	Food Safety Certification	1
or HSPM 2330	Hospitality Sanitation and Safety	
FSHN 4030	Food Laws and Regulations	2
FSHN 4200	Food Microbiology	3
Select 3 credits from the Food Microbiology area		
FSHN/MICRO 4070	Microbiological Safety of Foods of Animal Origins	
FSHN/MICRO 4210	Food Microbiology Laboratory	
Select 3 credits from the Food Processing area:		
FSHN 2070	Processing of Foods: Basic Principles and Applications	
FSHN 3050	Food Quality Management and Control	
ANS 2700 & 2700L	Foods of Animal Origin and Foods of Animal Origin Laboratory	
ANS 3600	Fresh Meat Science and Applied Muscle Biology	
FSHN 4710	Food Processing	
FSHN 4720	Food Processing Laboratory	

Minor in Food Science (Students majoring in Culinary Food Science cannot declare a minor in Food Science.)

FSHN 1010	Food and the Consumer	3
FSHN 2070	Processing of Foods: Basic Principles and Applications	3
Select 9 additional credits:		
Food chemistry:		
FSHN 3110	Food Chemistry (lab optional: FSHN 3110L)	3
FSHN 4100	Food Analysis	3
FSHN 4110	Food Ingredient Interactions and Formulations	2
Food microbiology:		
FSHN 4070	Microbiological Safety of Foods of Animal Origins	3
FSHN 4200	Food Microbiology	3
FSHN 4210	Food Microbiology Laboratory	3
Food processing/engineering:		
FSHN 3510	Introduction to Food Engineering Concepts	3
FSHN 4710	Food Processing	3
FSHN 4720	Food Processing Laboratory	2
General food science:		
FSHN 3050	Food Quality Management and Control	2
FSHN 4030	Food Laws and Regulations	2
FSHN 4060	Sensory Evaluation of Food	3

Minor in Nutrition (Students majoring in Diet and Exercise, Dietetics, and Nutritional Science cannot declare a minor in Nutrition.)

FSHN 1670	Introductory Human Nutrition and Health	3
FSHN 2650	Metabolism, Nutrition and Health	3
FSHN 3600	Advanced Nutrition and the Regulation of Metabolism in Health and Disease	3
Select at least 6 credits from:		

FSHN 2420	The US Food System	3
FSHN 2600X	Sports Nutrition	3
FSHN 2670	Clinical Perspectives on Nutrition	1
FSHN 3010	Precision Medicine in Integrated Health	1
FSHN 3610	Nutrition and Health Assessment	2
FSHN 3620	Nutrition and Health Throughout the Lifecycle	3
FSHN 3650	Obesity and Health	3
FSHN 4630	Community Nutrition and Health	3
FSHN 4670	Molecular Basis of Nutrition in Disease Etiology and Health Promotion	3
FSHN 4920	Research Concepts in Human Nutrition	2

Graduate Study

The Food Science and Human Nutrition (FSHN) Department offers coursework for the degrees master of science and doctor of philosophy with majors in food science and technology and in nutritional sciences, and minors in food science and technology and in nutrition. Graduate work in meat science is offered as a co-major in animal science and food science and technology.

Prerequisite to major work is a baccalaureate degree in food science, nutrition, or other physical or biological sciences or engineering that is substantially equivalent to those at Iowa State University.

Students taking major work for the degree doctor of philosophy either in food science and technology or in nutritional sciences may choose minors from other fields including anthropology, biorenewable resources and technology, chemistry, biochemistry, economics, education, journalism, microbiology, psychology, physiology, statistics, toxicology, or other related fields.

The Food Science and Technology (FST) graduate program offers MS and PhD degrees in the general areas of Food Chemistry and Functionality, Food Safety and Microbiology, and Food Processing. The FST core curriculum and interdisciplinary faculty team provides holistic graduate student training. Individuals with an undergraduate or graduate degree from a variety of academic training backgrounds, such as food sciences and the various disciplines of biology, chemistry, and agricultural sciences, may enter the FST program.

The interdepartmental graduate program in nutritional sciences, administered through the Graduate College, under the auspices of the Chairs of FSHN and Animal Science, will provide the structure for coordinating and enhancing interdisciplinary nutrition research and graduate education. Graduate students will be able to select from three specializations: animal nutrition, human nutrition, or molecular/biochemical nutrition. The three main departments are FSHN, Animal Science, and Kinesiology, whereas other departments (such as; Biochemistry, Biophysics, and Molecular Biology; Agronomy; and Statistics) may also be involved. (See Nutritional Sciences interdepartmental graduate major).

The Master of Professional Practice in Dietetics program is an online, course-work only, 12-month long, integrated graduate program that combines didactic coursework and on-site supervised experiential learning to train future Registered Dietitian Nutritionists. In addition to the required didactic coursework, students complete a minimum of 1000 hours of supervised experiential learning to meet the eligibility requirements to take the national credentialing exam for Registered Dietitian Nutritionists. Prerequisite for the program is graduation from a Didactic Program in Dietetics.

The department also offers an online 12-13 credit Graduate Certificate in Food Safety and Defense, in conjunction with the University of Nebraska, Lincoln, Kansas State University and the University of Missouri through the Great Plains Interactive Distance Education Alliance. Course topics include food microbiology, food defense, food toxicology, HACCP, and additional topics related to food safety. Students may be admitted if qualified for admission to the food science master's degree program.

The department offers work for concurrent B.S. and M.S. degree programs that allow students to obtain both the B.S. and M.S. degrees in 5 years. The programs are available to students majoring in nutritional science or pre-diet and exercise, and students progress toward M.S. degrees in nutritional sciences or diet and exercise, respectively. Students interested in these programs should contact the department for details. Application for admission to the Graduate College should be made during the junior year. Students begin research for the M.S. thesis or creative component during the summer after their junior year and are eligible for research assistantships.

Students graduating with advanced degrees in nutritional sciences and in food science and technology will demonstrate competency in their chosen discipline. Measurable outcomes will include the ability to:

- Apply scientific thinking to the analysis, synthesis and evaluation of knowledge within the discipline of food science, nutritional sciences, or dietetics
- Apply ethical reasoning within the discipline of food science, nutritional sciences or dietetics
- Effectively communicate discipline-specific information in written and oral forms to scientific audiences
- Effectively interact within scientific teams
- Facilitate learning within FSHN courses

Minors - Graduate

The department offers coursework for graduate minors in:

- food science/technology (https://fshn.hs.iastate.edu/graduate-students/graduate-programs/food-science-and-technology/)

- nutritional sciences (https://fshn.hs.iastate.edu/graduate-students/graduate-programs/interdepartmental-graduate-program-in-nutritional-sciences/)

Food Science and Technology Graduate Minor students must complete the following:

- 9 to 12 credits. Students without a background in food chemistry, food engineering/processing, and/or food microbiology are required to take FSHN 5110 and two 6000-level courses in two different competency areas.
- 9 credits of graduate level food science coursework as approved by the POS committee.
- Maximum of 3 credits at the 4000 level.

Nutritional Sciences Graduate Minor students must complete the following:

- 9 to 12 credits are required. Students who have not taken FSHN 3600 or its equivalent (advanced nutrition with a biochemistry perquisite) will need to take FSHN 3600, in which case the Nutrition Graduate minor will constitute 12 credits.
- 9 credits of graduate level nutrition courses as approved by the POS Committee.
- NUTRS 5100 and NUTRS 5110

Certificate - Undergraduate

Health Coach (https://kin.hs.iastate.edu/current-students/academics/health-coach-certificate/)

The undergraduate health coach certificate provides a rigorous academic and theoretical background in three components of health (nutrition, exercise and motivational coaching) required to prepare workers for the challenges of being a health coach.

Certificates - Graduate

Food Safety and Defense (http://www.fshn.hs.iastate.edu/graduate-program/food-safety-defense/)

The department offers an online 12-13 credit Graduate Certificate in Food Safety and Defense, in conjunction with the University of Nebraska, Lincoln, Kansas State University and the University of Missouri through the Great Plains Interactive Distance Education Alliance. Students may be admitted if qualified for admission to the food science master's degree program.

Courses primarily for undergraduates:

FSHN 1010: Food and the Consumer
Credits: 3. Contact Hours: Lecture 3.
The food system from point of harvest to the consumption of the food by the consumer. Properties of food constituents. Protection of food against deterioration and microbial contamination. Introduction of foods into the marketplace. Processes for making various foods. Government regulations. Use of food additives. Current and controversial topics. High school biology and chemistry or 3 credits of college level biology and chemistry recommended. (Typically Offered: Fall, Spring, Summer)

FSHN 1040: Introduction to Culinary Skills
Credits: 2. Contact Hours: Laboratory 4.
Prereq: (Credit or enrollment in HSPM 1330 *or* HSPM 2330*) or ServSafe® manager certification*
Introduction to culinary science. Students will develop hands-on culinary skills with a culinary professional. (Typically Offered: Spring)

FSHN 1100: Professional and Educational Preparation
Credits: 1. Contact Hours: Lecture 1.
Introduction to professional and educational development within the food science and human nutrition disciplines. Focus is on university and career acclimation as well as enhancement of communication skills. Offered on a satisfactory-fail basis only. (Typically Offered: Fall, Spring)

FSHN 1150: Food Preparation Laboratory
Credits: 1. Contact Hours: Laboratory 3.
Prereq: (Credit or enrollment in FSHN 1010 *OR FSHN 1110 OR* FSHN 2140*) and (Majors in Dietetics, Family and Consumer Sciences Education and Studies, Hospitality Management, Nutritional Science, and Diet & Exercise)*
Practice standard methods of food preparation with emphasis on quality, nutrient retention, and safety. (Typically Offered: Fall, Spring)

FSHN 1200: The Biochemistry of Beer
(Cross-listed with BBMB 1200).
Credits: 2. Contact Hours: Lecture 2.
An introduction to the major classes of biomolecules, basic biochemical concepts, enzymology, metabolism and genetic engineering as they apply to the production and flavor of beer. All aspects of the biochemistry of beer will be covered, including the malting of barley, starch conversion, yeast fermentation and the chemical changes that occur during the aging of beer. Intended for non-majors. (Typically Offered: Fall)

FSHN 1200L: Biochemistry of Beer Laboratory
(Cross-listed with BBMB 1200L).
Credits: 1. Contact Hours: Laboratory 3.
Prereq: Credit or enrollment in BBMB 1200
An introduction to biochemical methods related to the production of beer. Laboratory exercises related to water chemistry, mash enzymology, hop compound extraction and analysis, and yeast biology will be performed. Closely follows the material being taught in BBMB 1200. Graduation Restriction: Natural science majors are limited to elective credit only. (Typically Offered: Fall)

FSHN 1670: Introductory Human Nutrition and Health
Credits: 3. Contact Hours: Lecture 3.
Understanding and implementing present day knowledge of nutrition. The role of nutrition in the health and well being of the individual and family. High school biology or 3 credits of biology recommended. (Typically Offered: Fall, Spring, Summer)

FSHN 2030: Contemporary Issues in Food Science and Human Nutrition
Credits: 1. Contact Hours: Lecture 1.
Introduction to presentation of published research and discussion of current issues in food science and human nutrition. Emphasis on sources of credible information, ethics, and communication. (Typically Offered: Fall, Spring)

FSHN 2070: Processing of Foods: Basic Principles and Applications
Credits: 3. Contact Hours: Lecture 2, Laboratory 3.
Prereq: FSHN 1010
Lecture and lab-based instruction on principles of food processing and packaging. Food product-based discussion and activities will highlight raw food materials; unit operations; food quality and safety; processing plant sanitation; food forming and extrusion; fermentation; properties and selection of packaging materials. (Typically Offered: Spring)

FSHN 2140: Scientific Study of Food
Credits: 3. Contact Hours: Lecture 3.
Prereq: FSHN 1670 *or* FSHN 2650*;* CHEM 2310 *or* CHEM 3310*; plus concurrent enrollment in* FSHN 1150 *or 2150*
Composition and structure of foods. Principles of preparation of standard quality food products. Behavior and interactions of food constituents. (Typically Offered: Fall, Spring)

FSHN 2150: Advanced Food Preparation Laboratory
Credits: 2. Contact Hours: Laboratory 6.
Prereq: Credit or enrollment in FSHN 2140
Practice standard methods of food preparation with emphasis on quality, nutrient retention, and safety. Development of culinary skills and advanced food preparation. (Typically Offered: Spring)

FSHN 2200: American Food and Culture
Credits: 3. Contact Hours: Lecture 3.
American cuisine reflects the history of the U.S. It is the unique blend of diverse groups of people from around the world, including indigenous Native American Indians, Africans, Asians, Europeans, Pacific Islanders, and South Americans. Explore factors that impact the American Cuisine of today including diverse ethnic and cultural group influences, historical events related to food diversity in the U.S., and agriculture and industrial impacts on food production. Practical knowledge and basic food preparation techniques related to the U.S. food system and trends. Class sessions will include lectures, class discussions and Tasting Immersion activities. (Typically Offered: Fall, Spring)

FSHN 2420: The US Food System
Credits: 3. Contact Hours: Lecture 3.
Exploration of the components of our food system including production, processing, and access. A systems approach is used to evaluate the social, ethical, environmental, and nutrition/health implications of the US food system. Controversial topics related to how food is produced, processed, marketed and consumed will be discussed. (Typically Offered: Spring)

FSHN 2600X: Sports Nutrition
Credits: 3. Contact Hours: Lecture 3.
Principles of sports nutrition, emphasizing the role of nutrition in athletic performance, recovery, and health. Provides students with a foundational understanding of macronutrients, micronutrients, hydration, and supplementation as related to sports and exercise. (Typically Offered: Fall, Spring, Summer)

FSHN 2650: Metabolism, Nutrition and Health
Credits: 3. Contact Hours: Lecture 3.
Prereq: FSHN 1670 *and (credit or enrollment in* BBMB 2210*,* BBMB 3010*,* BBMB 3030*, or* BBMB 3160*)*
Fundamentals of nutrient metabolism and nutrient requirements. Role of macronutrient metabolism in physical performance and disease prevention. Effect of manipulation of macronutrient metabolism on physical performance and disease prevention. Applications of nutrient metabolism principles to dietary recommendations and planning. Current enrollment in 3000-level BBMB recommended. (Typically Offered: Spring, Summer)

FSHN 2670: Clinical Perspectives on Nutrition
Credits: 1. Contact Hours: Lecture 1.
Through case scenarios presented by practicing physicians and other health care professionals of various specializations, students will gain appreciation for the actual clinical impact of nutrition on health and disease. Through interactions with clinicians, students will learn about the current and future status of health care.

FSHN 2760: Grape and Wine Science
(Cross-listed with HORT 2760).
Credits: 3. Contact Hours: Lecture 4.
An introduction to viticulture (grape growing) and enology (wine making) to increase understanding of the science, including grape biology, chemistry, sensory of grape production and winemaking. Knowledge and the practical applications of grape growing and winemaking will be emphasized. No wine tasting. High school biology and chemistry or 3 credits of college biology and chemistry recommended.

FSHN 3010: Precision Medicine in Integrated Health
Credits: 3. Contact Hours: Lecture 3.
Role of personalized medicine in the prevention, diagnosis, and treatment of disease, with a focus on integrating genetic and genomic data into patient care. Application of pharmacogenomics, genomics, nutrigenomics, and nutrigenetics to inform clinical decision-making across a variety of health professions. Clinical pharmacogenetic guidelines and the application of genomic data in patient care will be discussed through interdisciplinary applications in lecture and case-based discussions. Health professional students will gain the knowledge and skills necessary to understand personalized medicine approaches in a collaborative, patient-centered healthcare environment. (Typically Offered: Fall)

FSHN 3050: Food Quality Management and Control
Credits: 2. Contact Hours: Lecture 2.
Prereq: 3 credits in statistics
Fundamentals of statistical decision-making processes and quality control procedures used in food quality assurance programs. (Typically Offered: Spring)

FSHN 3110: Food Chemistry
Credits: 3. Contact Hours: Lecture 3.
Prereq: CHEM 2310 *or* CHEM 3310
The structure, properties, and chemistry of food constituents and animal and plant commodities. Credit or enrollment in BBMB course 3000 or higher recommended. (Typically Offered: Fall)

FSHN 3110L: Food Chemistry Laboratory
Credits: 1. Contact Hours: Laboratory 3.
Prereq: Credit or concurrent enrollment in FSHN 3110
The laboratory practices of structure, properties, and chemistry of food constituents. (Typically Offered: Fall)

FSHN 3140: Professional Development for Culinary Food Science and Food Science Majors
Credits: 1. Contact Hours: Lecture 1.
Prereq: Major or minor in Culinary Food Science or Food Science; Junior or senior classification.
Introduction to the roles culinary scientists and food scientists hold within industry. Discussions focused on professional and educational development and emerging issues and trends in the food industry. (Typically Offered: Fall)

FSHN 3150: Professional Skills for Culinary Food Science and Food Science Majors
Credits: 1. Contact Hours: Lecture 1.
Prereq: Major or minor in FSHN
Focus on the importance of professional skills and application of those skills to potential job situations. Professional skills include communication, team building, leadership vs. management styles, business ethics, and continual learning. Junior classification recommended. (Typically Offered: Fall)

FSHN 3200: Taste of the Middle East and North Africa
(Cross-listed with ARABC 3200).
Credits: 3. Contact Hours: Lecture 3.
Exploration of Middle Eastern and North African food cultures through historical, religious, and social lenses. Students will engage in discussions, presentations, and cooking demonstrations and tastings. Taught in English. Meets International Perspectives Requirement. (Typically Offered: Fall, Spring)

FSHN 3400: Foundations of Dietetic Practice
Credits: 2. Contact Hours: Lecture 2.
Prereq: Major in Dietetics (A or H) or Pre-Diet and Exercise; Junior classification
Introduction to the profession of dietetics and responsibilities associated with dietetic professional practice. Emphasis on exploring career options in dietetics and preparation for supervised practice and graduate school. Leadership and professional career development for the dietitian is addressed through self reflection and creation of materials for post-baccalaureate programs. Professional issues related to dietetic practice include Code of Ethics, legal credentialing and standards of professional practice, leadership and future trends in the profession. (Typically Offered: Fall)

FSHN 3420: World Food Issues: Past and Present
(Cross-listed with ENVS 3420/ AGRON 3420).
Credits: 3. Contact Hours: Lecture 3.
Prereq: Junior classification
Issues associated with global agricultural and food systems including ethical, social, economic, environmental, and policy contexts. Investigation of various causes and consequences of overnutrition/ undernutrition, global health, poverty, hunger, access, and distribution. Meets International Perspectives Requirement. (Typically Offered: Fall, Spring, Summer)

FSHN 3510: Introduction to Food Engineering Concepts
Credits: 3. Contact Hours: Lecture 3.
*Prereq: (*FSHN 2070*; [*MATH 1600 *or* MATH 1650*]; [*PHYS 1150*,* PHYS 1310*, or* PHYS 2310*])*
Methodology for solving problems in food processing and introduction to food engineering concepts including food properties, material and energy balances, sources of energy, thermodynamics, fluid flow, heat transfer, and mass transfer. (Typically Offered: Spring)

FSHN 3600: Advanced Nutrition and the Regulation of Metabolism in Health and Disease
Credits: 3. Contact Hours: Lecture 3.
Prereq: ENGL 2500*;* FSHN 2650
Physiological and biochemical basis for nutrient needs; assessment of nutrient deficiency and toxicity; examination of nutrient functions and the regulation of metabolism; nutrient-gene interactions; mechanistic role of nutrients in health and disease. 3 credits or concurrent enrollment in 3000-level or above Biochemistry recommended. 3 credits in physiology recommended. (Typically Offered: Fall)

FSHN 3610: Nutrition and Health Assessment
Credits: 2. Contact Hours: Lecture 2, Laboratory 2.
Prereq: FSHN 2650
The assessment of nutritional status in individuals with application of the nutrition care process. Laboratory experiences in dietary, anthropometric, biochemical, and clinical assessment of nutritional status. . (Typically Offered: Spring)

FSHN 3620: Nutrition and Health Throughout the Lifecycle
Credits: 3. Contact Hours: Lecture 3.
Prereq: FSHN 3600*; credit or enrollment in* BIOL 2560 *or 3350*
Molecular, biochemical and physiological basis to understand the nutritional aspects of human development and aging. Nutrient needs and various disease states at each stage of human life cycle. (Typically Offered: Spring)

FSHN 3640: Nutrition and Prevention of Chronic Disease
Credits: 3. Contact Hours: Lecture 3.
*Prereq: (*BIOL 2560 *or* BIOL 3350*) or enrolled in NRS major*
Overview of nutrients, their functions, metabolism, food sources and optimal choices for the promotion of health and wellness. Nutrition strategies for the prevention of chronic disease, including cancer, diabetes and obesity, as they apply to individuals or the wider population will be discussed. (Typically Offered: Fall)

FSHN 3650: Obesity and Health
Credits: 3. Contact Hours: Lecture 3.
Multifactorial aspects of obesity, maintenance of healthy weight, and the relationship of weight status and chronic disease prevention. Traditional and novel nutrition and exercise theories as well as current popular diet and exercise trends will be discussed. (Typically Offered: Spring)

FSHN 3670: Medical Terminology for Health Professionals
Credits: 1. Contact Hours: Lecture 1.
An independent course focused on medical terminology, abbreviations, and simple clinical mathematical calculations. (Typically Offered: Fall, Spring, Summer)

FSHN 3730: Science and Practice of Brewing
(Cross-listed with ME 3730).
Credits: 3. Contact Hours: Lecture 1.5, Laboratory 4.5.
Prereq: 21 years of age and one course from this list: BIOL 2110*,* BIOL 2120*, CHEM 167,* CHEM 1770*,* PHYS 2310*,* PHYS 2320
Introduction to brewing science and technology. Understanding the role of malts, hops, water, and yeast in production of ale and lager beers. Unit operations in brewing. Health, safety, and environmental sustainability in alcohol production and consumption. Weekly laboratory in practical aspects of beer production. (Typically Offered: Fall, Spring)

FSHN 3800: Food Production Management
(Cross-listed with HSPM 3800).
Credits: 3. Contact Hours: Lecture 3.
Prereq: HSPM 1330 *or 2 cr MICRO;* FSHN 1010*, FSHN 1110 or* FSHN 2140*;* FSHN 1150 *or* FSHN 2150*; at least junior classification; enrollment in* HSPM 3800L
Principles of and procedures used in food production management including menu planning, costing, work methods, food production systems, quality control, and service. (Typically Offered: Fall, Spring)

FSHN 3800L: Food Production Management Experience
(Cross-listed with HSPM 3800L).
Credits: 3.
Prereq: HSPM 1330 *or 2 cr MICRO; FSHN 1110 or* FSHN 1010 *or* FSHN 2140*;* FSHN 1150 *or* FSHN 2150*; at least junior classification; enrollment in* HSPM 3800
Application of food production and service management principles and procedures in the program's foodservice operation. Field trip. (Typically Offered: Fall, Spring)

FSHN 3920: Food and Nutrition Services Management
Credits: 3. Contact Hours: Lecture 3.
Prereq: HSPM 3800*;* HSPM 3800L
Functions and responsibilities related to the management of foodservice systems and nutrition services, including planning, marketing, human resource management, and cost accounting. Accounting recommended.Graduation Restriction: Only one of HSPM 3920 or FSHN 3920 may count toward graduation. (Typically Offered: Spring)

FSHN 4030: Food Laws and Regulations
Credits: 2. Contact Hours: Lecture 2.
Prereq: FSHN 1010
History of food law in the US and the world. Relationship between policy, legislation and regulation. Introduction to primary US regulatory agencies and enforcement principles. Discussion of key laws related to food safety and nutrition. Overview of federal and independent research tools and sources of food law. (Typically Offered: Spring)

FSHN 4060: Sensory Evaluation of Food
(Dual-listed with FSHN 5060).
Credits: 3. Contact Hours: Lecture 2, Laboratory 3.
Prereq: FSHN 3050 *and credit or enrollment in* FSHN 4110*; 3 credits in statistics*
Sensory evaluation techniques used to evaluate the appearance, aroma, flavor, texture and acceptability of foods. Relationships between sensory and instrumental measurements of color and texture. Work independently and cooperatively (in a team) to identify sensory evaluation objectives, write hypotheses, design and conduct experiments, and analyze and interpret data. (Typically Offered: Fall)

FSHN 4070: Microbiological Safety of Foods of Animal Origins
(Dual-listed with FSHN 5070/ MICRO 5070). (Cross-listed with MICRO 4070).
Credits: 3. Contact Hours: Lecture 3.
*Prereq: (*MICRO 2010 *or* MICRO 3020*) or Instructor Permission*
Examination of the various factors in the production of foods, from production through processing, distribution and final consumption which contribute to the overall microbiological safety of the food. Upon successful completion of this class, the student will receive both the Preventive Controls for Human Foods certificate (FDA program) and the International HACCP Alliance certificate (USDA-FSIS program). Recommended: FSHN 4200 or MICRO 4200 and one semester of Microbiology Laboratory. (Typically Offered: Fall, Spring)

FSHN 4080: Dairy Products Evaluation
Credits: 1. Contact Hours: Laboratory 3.
Gain experience in identifying quality defects in dairy products including milk, cottage cheese, cheddar cheese, strawberry yogurt, butter, and vanilla ice cream. Intensive training for the National Collegiate Dairy Products Evaluation competition and for dairy product evaluation in the food industry. (Typically Offered: Spring)

FSHN 4100: Food Analysis
Credits: 3. Contact Hours: Lecture 2, Laboratory 3.
Prereq: FSHN 3110 *or* CHEM 2110
An introduction to the theory and application of chemical and instrumental methods for determining the constituents of food. Use of standard procedures for food analysis and food composition data bases. (Typically Offered: Fall)

FSHN 4110: Food Ingredient Interactions and Formulations
Credits: 2. Contact Hours: Lecture 1, Laboratory 3.
Prereq: FSHN 2140 *or* FSHN 3110
Application of food science principles to ingredient substitutions in food products. Laboratory procedures for standard formulations and instrumental evaluation, with emphasis on problem-solving and critical thinking. (Typically Offered: Fall, Spring)

FSHN 4120: Food Product Development
(Dual-listed with FSHN 5120).
Credits: 3. Contact Hours: Lecture 1, Laboratory 6.
Prereq: FSHN 4110*; senior classification*
Principles of developing consumer packaged food products. Application of skills gained in food chemistry, formulation, quality, sensory and processing. Emphasis on teamwork and effective communication. (Typically Offered: Spring)

FSHN 4190: Foodborne Hazards
(Cross-listed with MICRO 4190/ TOX 4190).
Credits: 3. Contact Hours: Lecture 3.
Prereq: MICRO 2010 *or* MICRO 3020*; 3 credits in BBMB*
Pathogenesis of human microbiological foodborne infections and intoxications, principles of toxicology, major classes of toxicants in the food supply, governmental regulation of foodborne hazards. Assessed service-learning component. Offered even-numbered years. Graduation Restriction: Only one of FSHN 4190 and FSHN 5190 may count toward graduation. (Typically Offered: Spring)

FSHN 4200: Food Microbiology
(Cross-listed with MICRO 4200/ TOX 4200).
Credits: 3. Contact Hours: Lecture 3.
Prereq: MICRO 2010 *or* MICRO 3020
Effects of microbial growth in foods. Methods to control, detect, and enumerate microorganisms in food and water. Foodborne infections and intoxications. (Typically Offered: Fall)

FSHN 4210: Food Microbiology Laboratory
(Cross-listed with MICRO 4210).
Credits: 3. Contact Hours: Lecture 1, Laboratory 5.
Prereq: MICRO 2010 *or* MICRO 3020*;* MICRO 3000L*. Credit or enrollment in FSHN/*MICRO 4200
Standard techniques used for the microbiological examination of foods. Independent and group projects on student-generated questions in food microbiology. Emphasis on oral and written communication and group interaction. (Typically Offered: Spring)

FSHN 4300: U.S. Health Systems and Policy
(Dual-listed with FSHN 5300).
Credits: 2. Contact Hours: Lecture 2.
Prereq: Junior, Senior, or Graduate Standing
Introduction to public policy for health care professionals. Emphasis on understanding the role of the practitioner for participating in the policy process, interpreting government policies and programs such as Medicare and Medicaid, determining reimbursement rates for eligible services, and understanding licensure and accreditation issues. Discussion and exploration of federal, state and professional policy-relevant resources. (Typically Offered: Fall, Spring)

FSHN 4350: Analysis of Food Markets
(Cross-listed with ECON 4350).
Credits: 3. Contact Hours: Lecture 3.
Prereq: ECON 3010 *and STAT 2260*
Food market analysis from an economics perspective; food markets and consumption; methods of economic analysis; food industry structure and organization; food and agriculture regulations; labeling; consumer concerns; agricultural commodity promotion. Final project required. (Typically Offered: Spring)

FSHN 4420: Food, Culture, and Sustainability
Credits: 3. Contact Hours: Lecture 3.
Prereq: Sophomore classification
Sustainable food, social, and health systems depend on positive interactions between culture, biology, and the environment. Historical and current cultural characteristics, such as common foods, meal patterns, food taboos, therapeutic uses of food, religion, immigration, and food acquisition patterns will be studied. Cultural acceptability, socio-economic equity, and ways to leverage healthy behaviors will be discussed. Meets U.S. Cultures and Communities Requirement. (Typically Offered: Fall)

FSHN 4600: Global Nutrition, Health and Sustainability
(Dual-listed with FSHN 5600).
Credits: 3. Contact Hours: Lecture 3.
Nutrition, health, and agriculture related issues in low- and middle-income countries are discussed using the framework of the Sustainable Development Goals. Topics include malnutrition, climate change, food insecurity/famine, micronutrient deficiencies, food production, the nutrition transition, water and sanitation challenges, and chronic diseases to enhance awareness of diverse worldviews and cultural humility. Systems thinking and the interaction of culture, biology, and the environment are emphasized. Meets International Perspectives requirement. Meets International Perspectives Requirement. (Typically Offered: Spring)

FSHN 4610: Medical Nutrition and Disease I
(Dual-listed with NUTRS 5610).
Credits: 4. Contact Hours: Lecture 4.
*Prereq: (Dietetics, Diet and Exercise, and Nutritional Science Majors) AND (*BIOL 2560 *or* BIOL 3350*), (*FSHN 3600*,* FSHN 3610*, and* FSHN 3670*)*
Pathophysiology of selected chronic disease states and their associated medical problems. Specific attention will be directed to medical nutrition needs of patients in the treatment of each disease state to optimize nutritional status and improve health. (Typically Offered: Fall)

FSHN 4630: Community Nutrition and Health
(Dual-listed with NUTRS 5630).
Credits: 3. Contact Hours: Lecture 3.
Prereq: FSHN 3610
Dual listed with FSHN 4630. Survey of current public health nutrition problems among nutritionally vulnerable individuals and groups. Discussion of the multidimensional nature of those problems and of community programs addressing them. Grant writing as a means for funding community nutrition program development. Significant emphasis on written and oral communication at the lay and professional level. Meets U.S. Cultures and Communities Requirement. (Typically Offered: Fall, Spring)

FSHN 4640: Medical Nutrition and Disease II
(Dual-listed with NUTRS 5640).
Credits: 4. Contact Hours: Lecture 4.
Prereq: FSHN 4610 *or* NUTRS 5610
Dual listed with FSHN 4640. Pathophysiology of selected acute and chronic disease states and their associated medical problems. Specific attention will be directed to medical nutrition needs of patients in the treatment of each disease state to optimize nutritional status and promote health. (Typically Offered: Fall, Spring, Summer)

FSHN 4660: Nutrition Counseling and Education Methods
(Dual-listed with FSHN 5660).
Credits: 3.
Prereq: FSHN 3610 *and* FSHN 3620
Application of counseling and learning theories with individuals and groups in community and clinical settings. Includes discussion and experience in building rapport, assessment, diagnosis, intervention, monitoring, evaluation, and documentation. Literature review of specific counseling and learning theories. (Typically Offered: Fall, Spring)

FSHN 4670: Molecular Basis of Nutrition in Disease Etiology and Health Promotion
Credits: 3. Contact Hours: Lecture 3.
Prereq: BBMB course 2000 or higher or instructor permission
Understanding the molecular basis for the role of nutrients, nutrient-derivatives, and bioactive compounds in the development, prevention, and treatment of common diseases including diabetes, cancer, vascular disease, obesity, neurological disease, birth defects, aberrant mineral metabolism, bone health, and autoimmune disease. Translating this understanding into practical approaches for improving the health of individuals and populations. Credit for FSHN 3600 suggested. (Typically Offered: Spring)

FSHN 4700: Sustainable and Healthy Eating Patterns
(Dual-listed with FSHN 5700).
Credits: 3. Contact Hours: Lecture 3.
Prereq: FSHN 1670 *or* FSHN 2420
Utilize a systems-based approach to consider health implications, economic considerations, environmental impacts, and sociocultural wellbeing to comprehensively explore sustainable diets. Evaluate benefits and challenges of transitioning to sustainable diets in the context of current and future food systems. Address the role of food environments, connect dietary guidelines to sustainability issues, and identify how sustainable diets fit in the context of global environmental goals.
Food systems frameworks and models will be applied to evaluate the sustainability of select foods and eating patterns at the local and global level. . (Typically Offered: Spring)

FSHN 4710: Food Processing
Credits: 3. Contact Hours: Lecture 3.
Prereq: (AE 4510 or CHE 3570 *or* FSHN 3510*)*
Principles and application of food processing using both thermal (ex., blanching, pasteurization, canning, drying, freezing, evaporation, aseptic processing, extrusion) and non-thermal (ex., high pressure, irradiation, Ohmic, pulsed electric field) unit operations. Emphasis on microbial inactivation for food safety and preservation, process heat and mass balance, and related numerical problem solving. (Typically Offered: Fall)

FSHN 4720: Food Processing Laboratory
Credits: 2. Contact Hours: Lecture 1, Laboratory 3.
Prereq: FSHN 4710
Hands-on and demonstration laboratory activities related to food processing principles and applications using lab and pilot-scale equipment. Laboratory experiences include important food processing operations, e.g., blanching/pasteurization, canning, freezing, drying, corn wet milling, fermentation, baking etc. Emphasis on mass balance, interpreting data, writing reports, and presentations. Occasional field trips. (Typically Offered: Spring)

FSHN 4760X: Science and Practice of Grape and Wine
(Dual-listed with FSHN 5760X).
Credits: 3. Contact Hours: Lecture 2, Laboratory 2.
Prereq: CHEM 1670 *or* CHEM 1770 *or* FSHN 1010 *or* BIOL 2110 *or BIOL 21210 or FSHN/*HORT 2760
Develop an understanding of complex concepts on the importance of both viticulture and enology in the final quality of wine. Students will learn about various topics from ISU faculty as well as guest speakers. This course would provide knowledge and tools to be applied in the future on grape growing, on grape and wine chemistry, on the process of winemaking, on how to evaluate a food and beverage product, and on how to make decisions based on the knowledge. . (Typically Offered: Spring)

FSHN 4770X: Dairy Products and Dairy Processing
(Dual-listed with FSHN 5770X).
Credits: 3. Contact Hours: Laboratory 3, Lecture 2.
*Prereq: Credit or concurrent enrollment in (*MICRO 2010 *or* MICRO 3020*) and (*CHEM 1630 *or* CHEM 1670 *or* CHEM 1770*)*
In-depth exploration of dairy chemistry, microbiology, and advanced processing technologies in the dairy industry. Topics include milk composition, quality assurance, microbial management, the production of dairy products such as cheese, yogurt, butter, ice cream, and powdered milk, and sensory evaluation. Traditional and emerging processing techniques, including thermal processing, homogenization, spray drying, membrane filtration, and novel dairy processing technologies will be discussed. The course also covers dairy plant management, sustainability, and the role of automation in modern dairy production, equipping students with the expertise needed for innovation in the evolving dairy and food sector. (Typically Offered: Spring)

FSHN 4820X: Global Cuisines
Credits: 3.
*Prereq: FSHN/*HSPM 3800*, FSHN/*HSPM 3800L*, and* FSHN 1040
Practice culinary techniques and food safety, analyze ingredients, and gain an understanding of culinary history within culinary traditions. Explore Asian, European, Latin and Middle Eastern cuisines emphasizing ingredient functionality, nutritional analysis, cooking methods, and cultural significance. Experience cooking demonstrations, hands-on cooking labs, lecture, a food innovation project, and sensory evaluation. (Typically Offered: Fall)

FSHN 4870: Fine Dining Management
(Dual-listed with HSPM 5870/ FSHN 5870). (Cross-listed with HSPM 4870).
Credits: 3. Contact Hours: Lecture 2, Laboratory 3.
Prereq: HSPM 3800*,* HSPM 3800L*, and* HSPM 1330*, or ServSafe® Certification*
Exploration of the historical and cultural development of the world food table. Creative experiences with U.S. regional and international foods. Application of management and financial principles in food preparation and service in fine dining settings. Meets International Perspectives Requirement. (Typically Offered: Fall)

FSHN 4900A: Independent Study: Dietetics
Credits: 1-6. Repeatable, maximum of 6 credits.
Prereq: Instructor Permission for Course
Independent work in food science, nutrition, or dietetics. Graduation Restriction: A maximum of 6 credits of FSHN 4900 may be used toward graduation. (Typically Offered: Fall, Spring, Summer)

FSHN 4900B: Independent Study: Food Science
Credits: 1-6. Repeatable, maximum of 6 credits.
Prereq: Instructor Permission for Course
Independent work in food science, nutrition, or dietetics. Graduation Restriction: A maximum of 6 credits of FSHN 4900 may be used toward graduation. (Typically Offered: Fall, Spring, Summer)

FSHN 4900C: Independent Study: Nutrition
Credits: 1-6. Repeatable, maximum of 6 credits.
Prereq: Instructor Permission for Course
Independent work in food science, nutrition, or dietetics. Graduation Restriction: A maximum of 6 credits of FSHN 4900 may be used toward graduation. (Typically Offered: Fall, Spring, Summer)

FSHN 4900D: Independent Study: International Experience
Credits: 1-6. Repeatable, maximum of 6 credits.
Independent work in food science, nutrition, or dietetics. Graduation Restriction: A maximum of 6 credits of FSHN 4900 may be used toward graduation. (Typically Offered: Fall, Spring, Summer)

FSHN 4900E: Independent Study: Entrepreneurship
Credits: 1-6. Repeatable, maximum of 6 credits.
Prereq: Instructor Permission for Course
Independent work in food science, nutrition, or dietetics.Graduation Restriction: A maximum of 6 credits of FSHN 4900 may be used toward graduation. (Typically Offered: Fall, Spring, Summer)

FSHN 4900H: Independent Study: Honors
Credits: 1-6. Repeatable, maximum of 6 credits.
Prereq: Permission of Instructor; Membership in the University Honors Program
Independent work in food science, nutrition, or dietetics. Graduation Restriction: A maximum of 6 credits of FSHN 4900 may be used toward graduation. (Typically Offered: Fall, Spring, Summer)

FSHN 4910A: Supervised Work Experience: Dietetics
Credits: 1-4. Repeatable, maximum of 4 credits.
Prereq: Permission of Instructor; Permission of Advisor
Supervised off-campus work experience relevant to the academic major. Graduation Restriction: No More Than 4 Credits Of FSHN 4910A,B,C May Be Applied Toward Graduation. Offered on a satisfactory-fail basis only. (Typically Offered: Fall, Spring, Summer)

FSHN 4910B: Supervised Work Experience: Food Science
Credits: 1-4. Repeatable, maximum of 4 credits.
Prereq: Permission of Instructor; Permission of Advisor
Supervised off-campus work experience relevant to the academic major. Graduation Restriction: A maximum of 4 credits of FSHN 4910 may be used toward graduation. Offered on a satisfactory-fail basis only. (Typically Offered: Fall, Spring, Summer)

FSHN 4910C: Supervised Work Experience: Nutrition
Credits: 1-4. Repeatable, maximum of 4 credits.
Prereq: Permission of Instructor; Permission of Advisor
Supervised off-campus work experience relevant to the academic major. Graduation Restriction: A maximum of 4 credits of FSHN 4910 may be used toward graduation. Offered on a satisfactory-fail basis only. (Typically Offered: Fall, Spring, Summer)

FSHN 4910D: Supervised Work Experience: Culinary Science
Credits: 1-4. Repeatable, maximum of 4 credits.
Prereq: Permission of Instructor; Permission of Advisor
Supervised off-campus work experience relevant to the academic major. Graduation Restriction: No More Than 4 Credits Of FSHN 4910A,B,C May Be Applied Toward Graduation. Offered on a satisfactory-fail basis only. (Typically Offered: Fall, Spring, Summer)

FSHN 4920: Research Concepts in Human Nutrition
Credits: 2.
Prereq: Junior standing or Permission of Instructor
Students will develop and implement research projects with faculty supervision, based on knowledge gained from nutrition, biology and chemistry courses and write a formal science paper to share the results of their research. Students will gain appreciation for independent research and experience creative and innovative aspects of nutrition research. (Typically Offered: Fall)

FSHN 4930: Food Preparation Workshop
Credits: 1-3. Repeatable, maximum of 3 credits.
Selected topics in food preparation including scientific principles, culture and culinary techniques. Variable format may include laboratory, recitation, and lecture. Offered on a satisfactory-fail basis only.

FSHN 4960A: Food Science and Human Nutrition Travel Course: International travel
(Dual-listed with FSHN 5960A).
Credits: 1-4. Repeatable.
(One credit per week traveled.) Limited enrollment. Tour and study of food industry, dietetic and nutritional agencies in different regions of the world. Pre-travel session arranged. Travel expenses paid by students. Meets International Perspectives Requirement. (Typically Offered: Fall, Spring, Summer)

FSHN 4960B: Food Science and Human Nutrition Travel Course: Domestic travel
(Dual-listed with FSHN 5960B).
Credits: 1-4. Repeatable.
(One credit per week traveled.) Limited enrollment. Tour and study of food industry, dietetic and nutritional agencies in different regions of the world. Pre-travel session arranged. Travel expenses paid by students. (Typically Offered: Fall, Spring, Summer)

FSHN 4980: Cooperative Education
Credits: Required. Repeatable, maximum of 0 credits.
Prereq: Department Chair Permission for Course
Required for students completing professional work periods in a cooperative education program. Students must register prior to commencing each work period. Offered on a satisfactory-fail basis only. (Typically Offered: Fall, Spring, Summer)

FSHN 4990: Undergraduate Research
Credits: 1-6. Repeatable, maximum of 6 credits.
Prereq: Instructor Permission for Course
Research under staff guidance. Graduation Restriction: A maximum of 6 credits of FSHN 4990 may be used toward graduation. (Typically Offered: Fall, Spring, Summer)

Courses primarily for graduate students, open to qualified undergraduates:

FSHN 5060: Sensory Evaluation of Food
(Dual-listed with FSHN 4060).
Credits: 3. Contact Hours: Lecture 2, Laboratory 3.
Prereq: FSHN 3050 *and credit or enrollment in* FSHN 4110*; 3 credits in statistics*
Sensory evaluation techniques used to evaluate the appearance, aroma, flavor, texture and acceptability of foods. Relationships between sensory and instrumental measurements of color and texture. Work independently and cooperatively (in a team) to identify sensory evaluation objectives, write hypotheses, design and conduct experiments, and analyze and interpret data. (Typically Offered: Fall)

FSHN 5070: Microbiological Safety of Foods of Animal Origins
(Dual-listed with FSHN 4070/ MICRO 4070). (Cross-listed with MICRO 5070).
Credits: 3. Contact Hours: Lecture 3.
Prereq: MICRO 4200 *or Graduate Classification*
Examination of the various factors in the production of foods, from production through processing, distribution and final consumption which contribute to the overall microbiological safety of the food. Upon successful completion of this class, the student will receive both the Preventive Controls for Human Foods certificate (FDA program) and the International HACCP Alliance certificate (USDA-FSIS program). Recommended: FSHN 4200 or MICRO 4200 and one semester of Microbiology Laboratory. (Typically Offered: Fall, Spring)

FSHN 5080: Consumer Perceptions and Nutrition Communication
Credits: 2. Contact Hours: Lecture 2.
Prereq: Acceptance in the Master of Professional Practice in Dietetics program
Examination of current consumer food and nutrition trends. Critical analysis of consumer perceptions relative to current research base. Use of various media to create effective nutrition messages for consumers. Activities designed to meet accreditation standards. (Typically Offered: Summer)

FSHN 5110: Integrated Food Science
Credits: 3. Contact Hours: Lecture 3.
Prereq: 3 credits in organic chemistry, physics, mathematics, and microbiology
Critical review of the key principles of food science and applications in the chemistry, microbiology, and processing of food. Understanding of the impact of processing on the quality of foods with respect to composition, quality and safety. (Typically Offered: Fall)

FSHN 5120: Food Product Development
(Dual-listed with FSHN 4120).
Credits: 3. Contact Hours: Lecture 1, Laboratory 6.
Prereq: FSHN 4110*; senior classification*
Principles of developing consumer packaged food products. Application of skills gained in food chemistry, formulation, quality, sensory and processing. Emphasis on teamwork and effective communication. (Typically Offered: Spring)

FSHN 5160: Advanced Nutrition I
Credits: 2. Contact Hours: Lecture 2.
Prereq: Acceptance in the Master of Professional Practice in Dietetics program
Study of the relationship between diet and health, focusing on the molecular and cellular mechanisms that underlie nutrient absorption, metabolism and disease risk. Contemporary issues in nutrition will be discussed. Activities designed to meet accreditation standards. (Typically Offered: Fall)

FSHN 5170: Gut Microbiome: Implications for Health and Diseases
(Cross-listed with ANS 5170/ MICRO 5170/ VMPM 5170).
Credits: 3. Contact Hours: Lecture 3.
Prereq: Graduate standing
Explore current research on gut microbiome including modern tools used to study the gut microbiome. Examine the linkages between gut microbiome and health status, diseases, and manipulation of gut microbiome to improve health. Basic knowledge in microbiology/ immunology is recommended. . (Typically Offered: Fall)

FSHN 5180: Advanced Nutrition II
Credits: 3. Contact Hours: Lecture 3.
Prereq: Acceptance in the Master of Professional Practice in Dietetics program
Principles of research design/methods and interpreting results/statistics in the current peer-reviewed scientific literature. Critical evaluation of the evidence-base to inform nutrition practice. Activities designed to meet accreditation standards. (Typically Offered: Spring)

FSHN 5210: Microbiology of Food
Credits: 2. Contact Hours: Lecture 2.
Prereq: A course in microbiology with laboratory; enrollment IDEA Food Safety and Defense Graduate Certificate or permission of instructor
Identification, enumeration, and characterization of bacteria, yeasts, and mold associated with foods and food processing. Effects of physical and chemical agents on micro-organisms will be studied. Microbiological problems in food spoilage, food preservation, food fermentation, and food-borne disease will be discussed. (Typically Offered: Spring, Summer)

FSHN 5220: Advanced Food Microbiology and Biotechnology
Credits: 2. Contact Hours: Lecture 2.
Prereq: Enrollment in IDEA Food Safety and Defense Graduate Certificate or permission of instructor
Basic principles in biotechnology and applied food microbiology, including current topics of interest in food biotechnology. Introduction to recombinant DNA techniques and how they are applied to genetically modify microorganisms, the use of nucleic acids as tools of rapid detection of microorganisms in foods, basic enzyme immobilization and down-stream processing techniques, and regulatory aspects of food biotechnology. Offered odd-numbered years. (Typically Offered: Spring).

FSHN 5230: A Multidisciplinary Overview of Food Safety and Security
Credits: 2. Contact Hours: Lecture 2.
Prereq: A course in biology or chemistry; enrollment in IDEA Food Safety and Defense Graduate Certificate or permission of instructor
Multidisciplinary food safety and security perspectives provided by numerous subject matter experts. Topics include food safety policy, ag bioterrorism, border security, animal ID, food defense and site security, risk analysis, crisis communication, epidemiology, HACCP, and more. (Typically Offered: Fall, Summer)

FSHN 5240: Food Microbiology
Credits: 3. Contact Hours: Lecture 3.
Prereq: A course in microbiology with laboratory; enrollment IDEA Food Safety and Defense Graduate Certificate or permission of instructor
Food Microbiology looks at the nature, physiology, and interactions of microorganisms in foods. The course is an introduction to food-borne diseases, the effect of food processing systems on the microflora of foods, principles of food preservation, food spoilage, and foods produced by microorganisms. Additionally, the course looks at food plant sanitation and criteria for establishing microbial standards for food products. (Typically Offered: Fall)

FSHN 5250: Principles of HACCP
Credits: 2. Contact Hours: Lecture 2.
Prereq: Enrollment in IDEA Food Safety and Defense Graduate Certificate or permission of instructor
A comprehensive study of the Hazard Analysis and Critical Control Point System and its application in the food industry. (Typically Offered: Fall)

FSHN 5260: Ethnic Foods: Food Safety, Food Protection and Defense
Credits: 2. Contact Hours: Lecture 2.
Prereq: Graduate standing; enrollment in IDEA Food Safety and Defense Graduate Certificate or permission of instructor
Understanding of the various factors that impact safety of ethnic and imported ethnic foods; knowledge about the handling, preparation, processing and storage of ethnic and imported foods and food products; science-based characterization of representative ethnic foods. (Typically Offered: Summer)

FSHN 5270: Microbiology of Fermented Foods
Credits: 2. Contact Hours: Lecture 2.
Prereq: Enrollment in IDEA Food Safety and Defense Graduate Certificate or permission of instructor
Microbiology of fermented foods covers the physiology, biochemistry, and genetics of microorganisms important in food fermentations. The course looks at how microorganisms are used in fermentations and the effects of processing and manufacturing conditions on production of fermented foods. (Typically Offered: Summer)

FSHN 5280: Food Protection and Defense-Essential Concepts
Credits: 2. Contact Hours: Lecture 2.
Prereq: Enrollment in IDEA Food Safety and Defense Graduate Certificate or permission of instructor
This course will provide students with an understanding of the principles required in a food defense program for a food manufacturing, warehousing or distribution center. The topics covered include: defining threats and aggressors; the Bioterrorism Act; food defense teams; vulnerability assessments; security programs; recall and traceability basics; security inspections; crisis management; emergency preparedness; and workplace violence. (Typically Offered: Spring)

FSHN 5290: Foodborne Toxicants
(Cross-listed with TOX 5290).
Credits: 2. Contact Hours: Lecture 2.
Mechanisms of action, metabolism, sources, remediation/detoxification, risk assessment of major foodborne toxicants of current interest, design of HAACP plans for use in food industries targeting foodborne toxicants, discussion of toxicants from a food defense perspective. Offered online only. (Typically Offered: Fall, Spring, Summer)

FSHN 5300: U.S. Health Systems and Policy
(Dual-listed with FSHN 4300).
Credits: 2. Contact Hours: Lecture 2.
Prereq: Junior, Senior, or Graduate Standing
Introduction to public policy for health care professionals. Emphasis on understanding the role of the practitioner for participating in the policy process, interpreting government policies and programs such as Medicare and Medicaid, determining reimbursement rates for eligible services, and understanding licensure and accreditation issues. Discussion and exploration of federal, state and professional policy-relevant resources. (Typically Offered: Fall, Spring)

FSHN 5330: Diet and Integrative Therapies for Prevention and Treatment of Diseases
Credits: 2. Contact Hours: Lecture 2.
Prereq: Acceptance in the Master of Professional Practice in Dietetics program
Explore the role of specific nutrients, dietary bioactive compounds and integrative therapies on disease prevention and treatment. Activities designed to meet accreditation standards. (Typically Offered: Fall)

FSHN 5370: Leadership and Management in Dietetics
Credits: 3. Contact Hours: Lecture 3.
Prereq: Acceptance in the Master of Professional Practice in Dietetics program
Application of leadership and management theories and approaches relevant to dietetics practice. Use of self-reflection and self-assessment to assist in recognition and development of leadership behaviors. Activities designed to meet accreditation standards. (Typically Offered: Summer)

FSHN 5380: Advanced Medical Nutrition Therapy
Credits: 3. Contact Hours: Lecture 3.
Prereq: Acceptance in the Master of Professional Practice in Dietetics program
Medical Nutrition Therapy for complex medical problems utilizing the scope and standards of dietetics practice, Nutrition Care Process, and practice-specific resources such as the Standards of Professional Performance. Evidence-based practice will be integrated into each topic. Capstone project. Activities designed to meet accreditation standards. (Typically Offered: Spring)

FSHN 5420A: Introduction to Molecular Biology Techniques: DNA Techniques
(Cross-listed with BMS 5420A/ EEOB 5420A/ BBMB 5420A/ GDCB 5420A/ HORT 5420A/ NREM 5420A/ NUTRS 5420A/ VDPAM 5420A/ VMPM 5420A).
Credits: 1. Contact Hours: Lecture 0.5, Laboratory 1.
Repeatable.
Prereq: Graduate Standing or Permission of Instructor
Includes genetic engineering procedures, sequencing, PCR, and genotyping. Offered on a satisfactory-fail basis only. (Typically Offered: Fall, Spring)

FSHN 5420B: Introduction to Molecular Biology Techniques: Protein
(Cross-listed with BMS 5420B/ EEOB 5420B/ BBMB 5420B/ GDCB 5420B/ HORT 5420B/ NREM 5420B/ NUTRS 5420B/ VDPAM 5420B).
Credits: 1. Repeatable.
Prereq: Graduate Standing or Permission of Instructor
Includes: immunophenotyping, ELISA, flow cytometry, microscopic techniques, image analysis, confocal, multiphoton and laser capture microdissection. Offered on a satisfactory-fail basis only. (Typically Offered: Spring, Summer)

FSHN 5420C: Introduction to Molecular Biology Techniques: Cell Techniques
(Cross-listed with BMS 5420C/ EEOB 5420C/ BBMB 5420C/ GDCB 5420C/ HORT 5420C/ NREM 5420C/ NUTRS 5420C/ VMPM 5420C/ VDPAM 5420C).
Credits: 1. Contact Hours: Laboratory 2.
Repeatable.
Prereq: Graduate Standing or Permission of Instructor
Includes: immunophenotyping, ELISA, flow cytometry, microscopic techniques, image analysis, confocal, multiphoton and laser capture microdissection.ular biology techniques and related procedures. Offered on a satisfactory-fail basis only. (Typically Offered: Fall, Spring)

FSHN 5420E: Introduction to Molecular Biology Techniques: Proteomics
(Cross-listed with BMS 5420E/ EEOB 5420E/ BBMB 5420E/ GDCB 5420E/ HORT 5420E/ NREM 5420E/ NUTRS 5420E/ VMPM 5420E/ VDPAM 5420E).
Credits: 1. Contact Hours: Lecture 0.5, Laboratory 1.
Repeatable.
Prereq: Graduate Standing or Permission of Instructor
Includes: two-dimensional electrophoresis, laser scanning, mass spectrometry, and database searching. Offered on a satisfactory-fail basis only. (Typically Offered: Fall)

FSHN 5420F: Introduction to Molecular Biology Techniques: Metabolomics
(Cross-listed with BMS 5420F/ EEOB 5420F/ BBMB 5420F/ GDCB 5420F/ HORT 5420F/ NREM 5420F/ NUTRS 5420F/ VMPM 5420F/ VDPAM 5420F).
Credits: 1. Contact Hours: Lecture 0.5, Laboratory 1.
Repeatable.
Prereq: Graduate Standing or Permission of Instructor
Includes: metabolomics and the techniques involved in metabolite profiling. For non-chemistry majoring students who are seeking analytical aspects into their biological research projects. Offered on a satisfactory-fail basis only. (Typically Offered: Fall)

FSHN 5540: Supervised Experience in Food Systems Management
Credits: 3.
Prereq: Acceptance in the Master of Professional Practice in Dietetics program
Combined supervised practice experiential and didactic learning course. Students demonstrate their competence of food service and management principles to ensure safe and efficient delivery of food and water to various populations; implement change management principles and demonstrate leadership in a variety of organizational settings. Experiences and course-related activities designed to meet accreditation standards. Capstone project. (Typically Offered: Summer)

FSHN 5550: Supervised Experience in Community Nutrition
Credits: 3.
Prereq: Acceptance in the Master of Professional Practice in Dietetics program
Combined supervised practice experiential and didactic learning course. Application of systems thinking and principles of sustainable, resilient, and healthy food and water systems to identify and assess nutrition problems in the community, design, implement and evaluate effectiveness of programs, and demonstrate competence in the relationship between policy, food, nutrition, health, and the environment. Experiences and course-related activities designed to meet accreditation standards. (Typically Offered: Fall)

FSHN 5560: Supervised Experience in Medical Nutrition Therapy
Credits: 5.
Prereq: Acceptance in the Master of Professional Practice in Dietetics program
Combined supervised practice experiential and didactic learning course. The Nutrition Care Process will be implemented in various populations, conditions and settings. Two weeks of staff relief at the end of the rotation. Experiences and course-related activities are designed to meet accreditation standards. (Typically Offered: Spring)

FSHN 5600: Global Nutrition, Health and Sustainability
(Dual-listed with FSHN 4600).
Credits: 3. Contact Hours: Lecture 3.
Prereq: Graduate Standing or Permission of Instructor
Nutrition, health, and agriculture related issues in low- and middle-income countries are discussed using the framework of the Sustainable Development Goals. Topics include malnutrition, climate change, food insecurity/famine, micronutrient deficiencies, food production, the nutrition transition, water and sanitation challenges, and chronic diseases to enhance awareness of diverse worldviews and cultural humility. Systems thinking and the interaction of culture, biology, and the environment are emphasized. Meets International Perspectives requirement. (Typically Offered: Spring)

FSHN 5620: Advanced Nutrition Assessment
Credits: 4. Contact Hours: Lecture 4.
Prereq: Acceptance in the Master of Professional Practice in Dietetics program
Overview and practical applications of methods for assessing nutritional status, including: theoretical framework of nutritional health and disease, dietary intake, biochemical indices, nutrition focused physical exam and body composition across the lifecycle. Activities designed to meet accreditation standards. (Typically Offered: Fall)

FSHN 5660: Nutrition Counseling and Education Methods
(Dual-listed with FSHN 4660).
Credits: 3.
Prereq: FSHN 3610 *and* FSHN 3620
Application of counseling and learning theories with individuals and groups in community and clinical settings. Includes discussion and experience in building rapport, assessment, diagnosis, intervention, monitoring, evaluation, and documentation. Literature review of specific counseling and learning theories. (Typically Offered: Fall)

FSHN 5700: Sustainable and Healthy Eating Patterns
(Dual-listed with FSHN 4700).
Credits: 3. Contact Hours: Lecture 3.
Prereq: Graduate classification
Utilize a systems-based approach to consider health implications, economic considerations, environmental impacts, and sociocultural wellbeing to comprehensively explore sustainable diets. Evaluate benefits and challenges of transitioning to sustainable diets in the context of current and future food systems. Address the role of food environments, connect dietary guidelines to sustainability issues, and identify how sustainable diets fit in the context of global environmental goals. Food systems frameworks and models will be applied to evaluate the sustainability of select foods and eating patterns at the local and global level. (Typically Offered: Spring)

FSHN 5750: Processed Foods
Credits: 3. Contact Hours: Lecture 3.
Prereq: FSHN 2140 *or* FSHN 3110*; A course in nutrition*
This course will examine effect of industrial and domestic food processing on the nutrient content of food and risk of developing chronic disease. Offered odd-numbered years. (Typically Offered: Spring).

FSHN 5870: Fine Dining Management
(Dual-listed with HSPM 4870/ FSHN 4870). (Cross-listed with HSPM 5870).
Credits: 3. Contact Hours: Lecture 2, Laboratory 3.
Prereq: Graduate Standing or Permission of Instructor
Exploration of the historical and cultural development of the world food table. Creative experiences with U.S. regional and international foods. Application of management and financial principles in food preparation and service in fine dining settings. Meets International Perspectives Requirement. (Typically Offered: Fall)

FSHN 5890: Systems Neuroscience: Brain, Behavior, and Nutrition-Related Integrative Physiology
(Cross-listed with NUTRS 5890/ GERON 5890/ NEURO 5890/ PSYCH 5890).
Credits: 2. Contact Hours: Lecture 2.
Prereq: Graduate Standing or Qualified Undergraduate with Permission of Instructor
Structural, functional, and biochemical aspects of brain and non-motor behavior across the human lifespan. Types of neuroimaging used to assess the brain. Current research is leveraged to gauge how nutrition, diseases related to nutrition, and associated physiological processes influence the brain, particularly for common developmental, psychological, and neurological disorders. (Typically Offered: Spring)

FSHN 5900A: Special Topics: Nutrition
Credits: 1-3. Repeatable, maximum of 6 credits.
Prereq: Instructor Permission for Course
(Typically Offered: Fall, Spring, Summer)

FSHN 5900B: Special Topics: Food Science
Credits: 1-3. Repeatable, maximum of 6 credits.
Prereq: Instructor Permission for Course
(Typically Offered: Fall, Spring, Summer)

FSHN 5900C: Special Topics: Teaching
Credits: 1-3. Repeatable, maximum of 6 credits.
Prereq: Instructor Permission for Course
(Typically Offered: Fall, Spring, Summer)

FSHN 5960A: Food Science and Human Nutrition Travel Course: International Travel
(Dual-listed with FSHN 4960A).
Credits: 1-4. Repeatable.
(One credit per week traveled.) Limited enrollment. Tour and study of food industry, dietetic and nutritional agencies in different regions of the world. Pre-travel session arranged. Travel expenses paid by students. Meets International Perspectives Requirement. (Typically Offered: Fall, Spring, Summer)

FSHN 5960B: Food Science and Human Nutrition Travel Course: Domestic travel
(Dual-listed with FSHN 4960B).
Credits: 1-4. Repeatable.
Prereq: Instructor Permission for Course
(One credit per week traveled.) Limited enrollment. Tour and study of food industry, dietetic and nutritional agencies in different regions of the world. Pre-travel session arranged. Travel expenses paid by students. (Typically Offered: Fall, Spring, Summer)

FSHN 5990: Creative Component
Credits: 1-30. Repeatable.
Prereq: Instructor Permission for Course
Nonthesis option only. (Typically Offered: Fall, Spring, Summer)

Courses for graduate students:

FSHN 6060: Advanced Food Analysis and Instrumentation
Credits: 3. Contact Hours: Lecture 2, Laboratory 3.
Prereq: FSHN 3110 *or* FSHN 4100 *or* FSHN 5110 *or equivalent*
Instrumental methods for measuring chemical and physical properties of foods, food quality and functionality. Techniques for methods development, sample preparation, optimization of operating conditions, and data analysis needed to obtain accurate, reproducible results by means of instrumentation. Offered even-numbered years. (Typically Offered: Fall). .

FSHN 6110: Advanced Food Processing
Credits: 3. Contact Hours: Lecture 3.
Recent advances in the science and technology of food processing and preservation; examples include both thermal and non-thermal processes, including cold plasma, nanotechnology, and extrusion. Advances in extraction and separation technologies, by-product utilization, and sustainability in food processing industry will also be discussed. Students to research on select topics and present. Offered odd-numbered years. (Typically Offered: Fall).

FSHN 6120: Advanced Food Chemistry
Credits: 3. Contact Hours: Lecture 3.
Prereq: FSHN 3110 *or* FSHN 4110 *or* FSHN 5110 *or* BBMB 4040 *or equivalent*
Structure, chemical and physical properties of lipids, proteins and carbohydrates, and their food and industrial applications. Changes in functionalities during processing and storage. Offered even-numbered years. (Typically Offered: Spring).

FSHN 6260: Advanced Food Microbiology
(Cross-listed with MICRO 6260/ TOX 6260).
Credits: 3. Contact Hours: Lecture 3.
Topics of current interest in food microbiology, including new foodborne pathogens, rapid identification methods, effect of food properties and new preservation techniques on microbial growth, and mode of action of antimicrobials. Offered odd-numbered years. (Typically Offered: Spring)

FSHN 6530: Food and Agricultural Traceability
Credits: 3. Contact Hours: Lecture 3.
Prereq: Enrollment in the Ivy Executive MBA program within the Ivy College of Business
Current issues and concepts of food and agricultural product traceability in the U.S., from production to consumption. Food types, microbial agents of concern, adulterants, disease investigations, risk analysis, risk mitigation, prevention and regulatory policy and advocacy. Travel to Washington, D.C. (Typically Offered: Fall)

FSHN 6810: Seminar
Credits: 1. Contact Hours: Lecture 1.
Repeatable, maximum of 2 credits.
Presentation of thesis or dissertation research. Must be taken once for each graduate program; once for the M.S. program and once for the Ph.D. program. Offered on a satisfactory-fail basis only. Offered on a satisfactory-fail basis only. (Typically Offered: Fall, Spring, Summer)

FSHN 6820: Research Presentation and Discussion
Credits: 1. Contact Hours: Lecture 1.
Repeatable.
Development of presentation skills and critical thinking skills through the presentation and discussion of research within student's discipline. Required each semester for all FSHN graduate students. . Offered on a satisfactory-fail basis only. (Typically Offered: Fall, Spring)

FSHN 6900: Special Problems
Credits: 1-30. Repeatable.
Prereq: Instructor Permission for Course
(Typically Offered: Fall, Spring, Summer)

FSHN 6950: Grant Proposal Writing
(Cross-listed with NUTRS 6950).
Credits: 1. Contact Hours: Lecture 1.
Prereq: 3 credits of graduate course work in food science or nutritional sciences, or a biological science discipline
Grant proposal preparation experiences including writing and critiquing ofproposals and budget planning. Understanding the grant funding process from federal, foundation, and commodity agencies. Includes preparing a grant for possible submission and participation in the review of proposals. Discussion of the role of successful grant writing in career development. (Typically Offered: Fall)

FSHN 6990: Research in Food Science and Technology
Credits: 1-30. Repeatable.
Prereq: Instructor Permission for Course
Offered on a satisfactory-fail basis only. (Typically Offered: Fall, Spring, Summer)